Agenthon 2026

T2 Quant V3

PCA Yield-Curve Factor Model + Tuning Report

Core question	V2 showed that a simple correlation/t-copula replacement was not enough. Can we better preserve cross-tenor structure by decomposing the UST curve into Level/Slope/Curvature-like factors and simulating directly in factor space?

What this report should make clear

- Why PCA is a natural model for a Treasury yield curve
- How the PCA factor model was connected to the existing forecast engine
- Results from the 2-factor / 3-factor / residual-noise ablation
- The 12-combination PCA2 residual-scale x recent-mix tuning grid
- The regime instability revealed by a development / validation split
- Why the next step should be an Adaptive PCA model

Core V3 question	To improve the joint score, is it better to model the curve movement itself in a low-dimensional factor space rather than keep adjusting pairwise correlations?

1. Why PCA - a yield curve is low-dimensional

UST 2Y/5Y/10Y/30Y typically do not move independently. Recurrent patterns include a parallel shift of the whole curve, a change in the short-long spread, and relative movement concentrated in intermediate maturities.

Factor intuition	Yield-curve movement	Example
PC1 ≈ Level	All tenors move in the same direction	2Y/5Y/10Y/30Y all move about +10bp
PC2 ≈ Slope	Short and long maturities move differently	2Y +15bp, 30Y +3bp -> flattening
PC3 ≈ Curvature	Intermediate tenors move relatively more	5Y/10Y move more than the curve ends

Important nuance PC1/PC2/PC3 are not defined to be Level/Slope/Curvature by construction. Rather, the leading empirical eigenvectors often resemble those economically familiar shapes.

2. PCA equations - in QRM language

First, daily yield changes are standardized so each tenor has approximately zero mean and unit variance.

$$z_{i,t} = (\Delta y_{i,t} - \mu_i) / s_i$$

The correlation/covariance matrix of the standardized vector z_t is then decomposed into eigenvalues and eigenvectors.

$$R = V \Lambda V^T$$

Keeping the eigenvectors associated with the largest eigenvalues yields the dominant curve factors.

$$f_t = V_k^T z_t$$

$$z_t \approx V_k f_t + \varepsilon_t$$

With k=2, the model mainly retains Level- and Slope-like movements. With k=3, it also keeps a Curvature-like component. The residual ε_t captures tenor-specific movement not explained by the retained factors.

3. V3 forecast engine - PCA factor bootstrap

V3 kept the EWMA-blended marginal volatility from V1, because the ablation had already shown that it was useful. Only the dependence mechanism was moved into PCA factor space.

$$\Delta y_t^* = D_t \cdot \text{normalize}(V_k f_t^* + r\, \varepsilon_t^*)$$

Symbol	Meaning
D_t	Current regime-sensitive, EWMA-blended tenor volatility
V_k	PCA loading matrix
f*_t	Bootstrapped historical factor shock
r	residual_scale: 0 = factor-only; 0.25 = restore 25% of residual noise
ε*_t	Bootstrapped residual shock not explained by the retained PCA factors

Recent factor pool

$$P(\text{recent factor pool}) = \text{recent_mix}$$

As in V1, the model can emphasize the current regime. The difference is that the recent/full mixture is now applied to factor shocks rather than raw tenor shocks.

4. First V3 ablation - 2 factors versus 3 factors

After a 2-fold smoke test, the main comparison was run with 24 folds and 1,000 draws. Three PCA structures were tested.

Model	Structure	Purpose
pca2_resid25	2 factors + residual 25%	Level/Slope core plus some tenor-specific noise
pca3_resid25	3 factors + residual 25%	Adds a Curvature-like factor
pca3_factor_only	3 factors + residual 0%	Pure factor structure

24-fold result

Model	Marginal ↓	Joint ↓	Tail ↓	Composite ↓	Win rate vs Ref
Reference Gaussian	1.1858	4.4381	0.1579	1.9559	-
PCA2 + resid 25%	1.1405	4.6348	0.1284	1.9864	50.0%
PCA3 factor only	1.1464	4.6644	0.1307	1.9987	50.0%
Quant block1	1.1477	4.6657	0.1268	1.9989	41.7%
PCA3 + resid 25%	1.1464	4.6661	0.1306	1.9992	50.0%

Decision

PCA2 + residual 25% became the strongest Quant model so far. It improved both marginal and joint scores relative to block1. The third factor appeared to add more noise than useful structure.

5. What PCA2 improved

Comparison	Marginal	Joint	Tail	Composite
Reference Gaussian	1.1858	4.4381	0.1579	1.9559
Quant block1	1.1477	4.6657	0.1268	1.9989
PCA2 resid25	1.1405	4.6348	0.1284	1.9864

Interpretation

Component	PCA2 vs block1	PCA2 vs Reference
Marginal	About 0.6% better	About 3.8% better
Joint	About 0.7% better	About 4.4% worse
Tail	About 1.2% worse	About 18.7% better
Composite	About 0.6% better	About 1.6% worse

What changed? For the first time, the joint score improved in the intended direction even in the full 24-fold experiment. PCA2 still did not match the Reference in joint coherence, but the factor idea produced a real improvement.

6. V3.1 tuning - residual scale x recent mix

Because the 2-factor structure was strongest, the number of factors was fixed at two and only two parameters were tuned.

residual_scale in {0, 0.10, 0.25, 0.50}

recent_mix in {0, 0.25, 0.60}

This produced 12 configurations over the same 24-fold walk-forward sample. For example, pca2_r10_m60 means residual_scale = 0.10 and recent_mix = 0.60.

Top full-sample results

Model	Composite ↓	Marginal ↓	Joint ↓	Tail ↓	Win rate vs Ref
Reference Gaussian	1.9559	1.1858	4.4381	0.1579	-
pca2_r10_m60	1.9850	1.1409	4.6291	0.1291	50.0%
pca2_r00_m60	1.9852	1.1413	4.6290	0.1291	50.0%
pca2_r25_m60	1.9864	1.1405	4.6348	0.1284	50.0%
pca2_r50_m60	1.9864	1.1409	4.6343	0.1285	45.8%
quant_block1	1.9989	1.1477	4.6657	0.1268	41.7%

Full-sample result

Several PCA2 settings consistently beat block1, but none beat the Reference score of 1.9559. Making the grid finer would increase the risk of tuning to this same 24-fold sample.

7. Development / validation split - the more important finding

Selecting the best parameter on the same 24 folds used for evaluation risks overfitting. The folds were therefore split chronologically into 12 development folds and 12 validation folds to see whether a setting selected earlier would remain strong later.

Segment	Period	Best PCA2	Composite ↓	Reference ↓	Observation
Development	2022-08-24 to 2023-07-28	r00_m60	1.9923	2.0031	PCA2 narrowly beat Refe
Validation	2023-08-28 to 2024-08-01	r50_m25	1.9527	1.9088	The best configuration c
Dev choice on Validation	Same validation period	r00_m60	1.9781	1.9088	The dev-optimal setting

Key finding

There was no single fixed optimal residual scale and recent mix. Development favored residual = 0 and recent_mix = 60%, while Validation favored residual = 50% and recent_mix = 25%. This strongly suggests regime-dependent parameters.

8. Why fixed parameters moved - regime instability

recent_mix and residual_scale are really numerical statements about how different the current market is from the long-run historical distribution. It is therefore plausible that the same value should not be used at every forecast origin.

Market state	recent_mix direction	residual direction	Intuition
Recent volatility/correlation differs sharply from long run	Higher	Adaptive	Emphasize recent factor shocks
Recent regime resembles long-run average	Lower	Stable	Use more full-history information
Curve is explained well by common factors	-	Lower	Level/Slope factors are sufficient
Tenor-specific noise rises	-	Higher	Restore more PCA residual risk

Concept for the next model

$$\text{recent_mix}_t = f(\ \sigma_{recent} / \sigma_{long},\ ||R_{recent} - R_{long}||\ )$$

$$\text{residual_scale}_t = g(\ \text{recent PCA explained variance, tenor-specific residual risk}\ )$$

These rules must be deterministic and use only information available at the forecast origin. Choosing parameters after observing future outcomes would be leakage.

9. V3 final checklist

Question	Answer from V3
Why PCA?	UST joint dependence looked more like a yield-curve factor problem than a pure correlation prob
2 factors or 3 factors?	Two factors were more stable in the 24-fold test.
Did PCA improve on block1?	Yes. Best PCA2 composite = 1.9850 versus block1 = 1.9989.
Did it beat the Reference?	Not yet. Reference = 1.9559 on the full sample.
What happened in Development?	r00_m60 narrowly beat the Reference.
Did the same parameters remain best in Validation?	No. The best setting changed to r50_m25.
Most important new finding?	A fixed tuning choice may be less robust than an adaptive regime rule.

One-line conclusion

PCA improved the joint structure, but the preferred recent/residual weights changed across regimes. The conclusion of V3 is not to tune the grid more aggressively, but to move toward Adaptive PCA.

Next step

Quant V4 Adaptive PCA: determine recent_mix and residual_scale at each forecast origin from recent/long volatility ratios, correlation-regime distance, and PCA explained-variance diagnostics.